

Validation report: PD scorecard on Lending Club 36-month loans

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Headline (out-of-time 2014–2015). AUC 0.639, Gini 0.277, KS 0.199; calibration ratio 0.890; lifetime EL / realized loss 0.836; score PSI 0.003.

Scope

Independent (second-line) review of a probability-of-default scorecard for unsecured 36-month consumer loans. The target is a lifetime PD over the 36-month contractual term (not a 12-month regulatory PD). Development: vintages 2007–2013 (80/20 split by loan id); out-of-time validation: vintages 2014–2015, the latest vintages fully matured at the 2018Q4 data cut-off.

Overall opinion

Not fit for loss estimation in its current form. Acceptable for rank-ordering applicants, subject to recalibration before any use of PD levels.

Findings

1. **[High]** Calibration out of time. Mean PD 0.129 against an observed default rate of 0.145 (ratio 0.890, slope 0.95, largest decile gap 2.4 pp, Hosmer–Lemeshow $p < 1e-16$). The score population is stable, so the gap reflects a different default rate in the 2014–2015 vintages, not a change in the applicant mix. Estimates are non-conservative when the ratio is below one.
2. **[High]** Expected loss backtest. On the same exposure-at-default base, lifetime expected loss is 0.836 times realized loss out of time (predicted LGD 0.895 vs realized 0.891). LGD is close on average, so the difference is driven mainly by PD.
3. **[Medium]** Discrimination. Out-of-time AUC 0.639 against 0.661 for Lending Club’s grade and 0.661 for the XGBoost challenger on all features. Restricted to the scorecard’s features, XGBoost reaches 0.644: most of the challenger’s advantage comes from the extra features (total IV 0.39 for the scorecard), not from the non-linear form.
4. **[Low]** Stability. Score PSI 0.003. Variables flagged: `verification_status` 0.165 (monitor), `purpose` 0.106 (monitor).
5. **[Low]** LGD differentiation. Predicted LGD ranges only from 0.849 to 0.922: the LGD model works as a segment average rather than a ranking tool.

Model choice: accuracy versus explainability

The scorecard is additive in weight-of-evidence bins, so every point of the score can be traced to an applicant attribute and explained in an adverse-action notice. The challenger’s extra discrimination comes mostly from using more features; on the same features the gain is small, which supports keeping the interpretable form and investing in data rather than in model complexity.

Limitations

Approved applicants only (no reject inference); 60-month loans and vintages after 2015 excluded because they are immature; US platform data; exposure at default approximated as the funded amount times the average EAD/funded ratio of development defaults (0.586); LGD ignores discounting; the challenger used an early-stopping set carved out of the training sample, so the holdout remains independent.

Recommendations

1. Recalibrate the PD level on the most recent matured vintages (calibration-in-the-large, then slope) and re-run the binomial and Hosmer–Lemeshow tests before using PD levels.
2. Enrich the application data with bureau attributes available at origination (for example revolving balance-to-limit ratios and recent account openings), and re-assess discrimination.
3. Replace the constant exposure factor with an amortization-based EAD model.
4. Monitor the flagged variables and the score PSI quarterly with the thresholds 0.10 (monitor) and 0.25 (action).

Supporting tables

Discrimination: scorecard

sample	n	default_rate	auc	gini	ks
train	140225	0.126	0.651	0.301	0.220
holdout	35201	0.126	0.652	0.304	0.226
oot	445596	0.145	0.639	0.277	0.199

Discrimination: challenger (XGBoost)

sample	n	default_rate	auc	gini	ks
train	140225	0.126	0.695	0.390	0.282
holdout	35201	0.126	0.674	0.349	0.263
oot	445596	0.145	0.661	0.322	0.232

Calibration by PD decile (OOT)

group	n	defaults	observed_dr	mean_pd	binom_pvalue
1	44560	2048	0.046	0.045	0.271
2	44560	3544	0.080	0.069	0.000
3	44559	4445	0.100	0.085	0.000
4	44560	5262	0.118	0.099	0.000
5	44559	5967	0.134	0.113	0.000
6	44560	6780	0.152	0.129	0.000
7	44559	7407	0.166	0.145	0.000
8	44560	8225	0.185	0.165	0.000
9	44559	9476	0.213	0.191	0.000
10	44560	11293	0.253	0.246	0.000

Population stability (PSI)

variable	psi	status
loan_amnt	0.019	stable
annual_inc	0.009	stable
dti	0.066	stable
emp_length_years	0.023	stable
fico_mid	0.058	stable
inq_last_6mths	0.034	stable
delinq_2yrs	0.026	stable
pub_rec	0.001	stable
open_acc	0.034	stable
total_acc	0.014	stable
revol_bal	0.009	stable
revol_util	0.031	stable
credit_history_years	0.054	stable
home_ownership	0.015	stable
verification_status	0.165	monitor
purpose	0.106	monitor
score	0.003	stable

Expected loss backtest (OOT)

vintage	n	exposure	expected_loss	realized_loss	ratio
2014	162570	2,046,040,750	126,637,998	136,945,538	0.925
2015	283026	3,624,300,800	216,865,017	274,011,577	0.791

Expected loss sensitivity (OOT)

scenario	expected_loss
base	343,503,016
PD x1.2	412,203,619
LGD +10pp	381,613,188
PD x1.2 & LGD +10pp	457,935,826

Scorecard coefficients

feature	coef
fico_mid	-0.797
annual_inc	-0.825
revol_util	-0.351
dti	-0.583
purpose	-1.055
inq_last_6mths	-1.069
home_ownership	-0.419

Figures

